

Literature Benchmark Brief: Pellegrino_2021_TII

Title: A holistic approach for offshore wind farm SCADA monitoring and fault diagnosis

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Cluster: Cluster B: SCADA Degradation & Reliability

Core Theoretical Mechanism & Model Formulation

Multi-sensor SCADA anomaly isolation under asynchronous timestamping

Empirical Scale & Experimental Dataset

North Sea utility-scale offshore wind farm

Relevance to IEEE TSTE & Sustainable Energy Systems

Reveals high frequency of corrupted pitch and anemometer registers in operational SCADA

Identified Gaps & Operational Vulnerabilities

Focuses on component health monitoring rather than grid pre-dispatch reserve adequacy

Role in Manuscript Defense & Reviewer #2 Proof

Validates why blade-pitch channels are treated as vulnerable/withheld in utility environments